

2	(a)	Gravitational potential at a point is the <u>work done by an external force per unit mass</u> to move a <u>small point mass</u> from infinity to that point without a change in kinetic energy.	B2
	(b)	$\Phi = -\frac{GM_{\text{earth}}}{r_1} - \frac{GM_{\text{moon}}}{r_2}$ $\Phi = -\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{3.84 \times 10^8 - 1.74 \times 10^6} - \frac{6.67 \times 10^{-11} \times 7.35 \times 10^{22}}{1.74 \times 10^6}$ $= -3.86 \times 10^6 \text{ J kg}^{-1}$	C1 M1 A0
	(c)	(i) Calculation of the gradient Taking adjacent values $\text{gradient} = \frac{(-1.95 - (-1.97)) \times 10^6}{(2.01 - 1.99) \times 10^8}$ $= 0.010 \text{ N kg}^{-1}$	C1 A1
		(ii) Gravitational force provides centripetal force $mg = \frac{mv^2}{r}$ $v = \sqrt{gr}$ $v = \sqrt{(0.010)(2 \times 10^8 + 6.38 \times 10^6)}$ $= 1437 = 1440 \text{ m s}^{-1}$ (subtract one mark if the radius of the earth was not considered)	C1 A1
	(d)	The spacecraft requires at least enough kinetic energy to cross the maximum potential point at $3.40 \times 10^8 \text{ m}$ from the surface of the Earth. $KE_1 + PE_1 = KE_2 + PE_2$ $KE + m\Phi_1 = 0 + m\Phi_2$ $KE = 1000(-1.28 - (-62.5)) \times 10^6$ $= 6.12 \times 10^{10} \text{ J}$	M1 A1

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3	(a)	The internal energy of a system is the sum of a random distribution of	B1
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